DASC 3240, L17

R basics – Data manipulation and ggplot2

2023/24 SPRING SEMESTER

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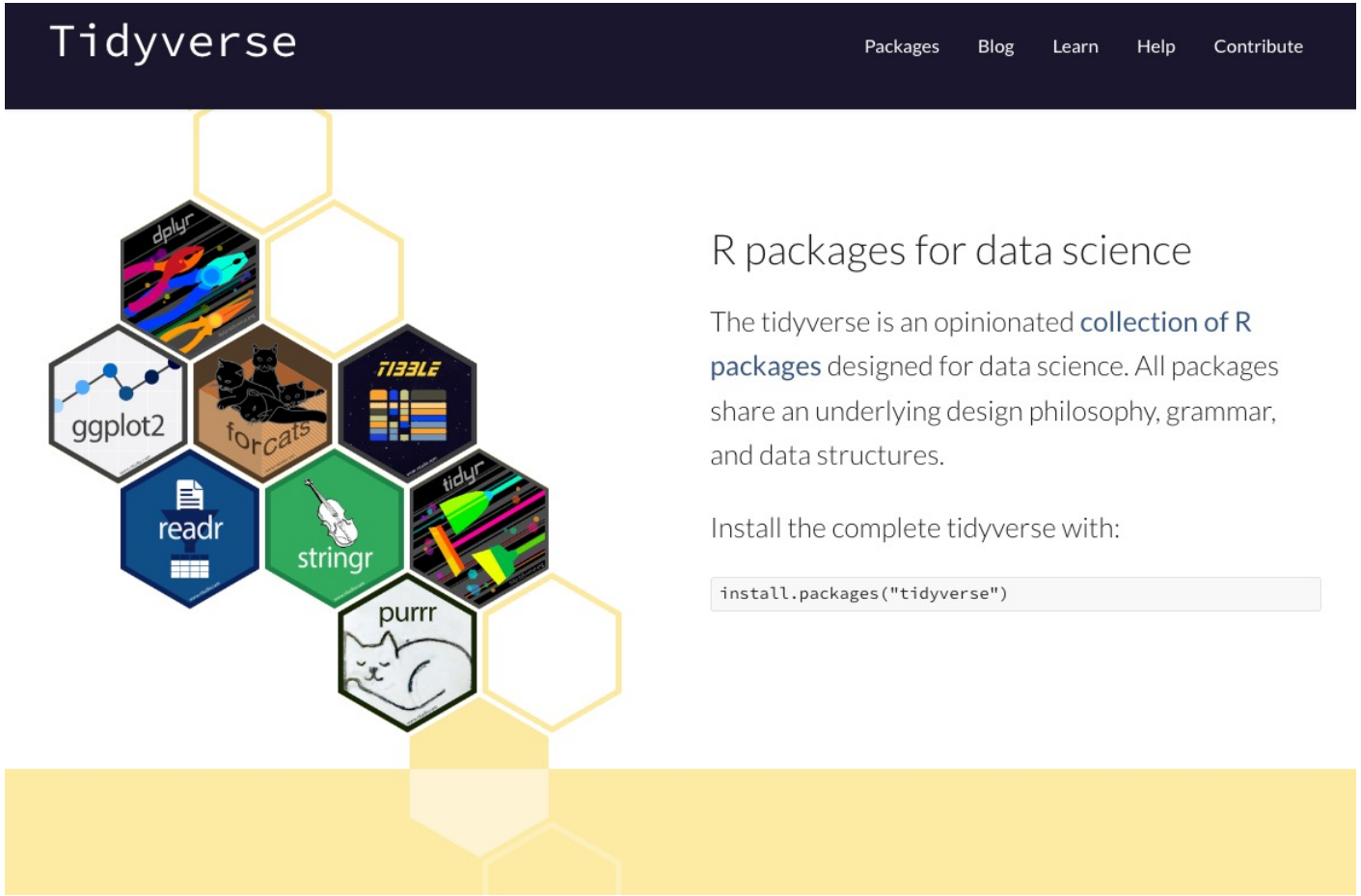
Outline

1. “Tidyverse” package
2. Data manipulation with tidyverse
3. “ggplot2” basics
4. Hands-on

“tidyverse” package

“tidyverse” package

- There are some commonly used packages (like “pandas”, “numpy”, “scipy”, and “matplotlib” in Python).
- “**tidyverse**” is one of them.
- It is a collection of several packages developed by **Hadley Wickham** and other contributors. “tidyverse” contains: **“ggplot2”**, **“dplyr”**, **“tidyr”**, **“readr”**, **“purrr”**, **“tibble”**, **“stringr”**, **“forcats”**, **“lubridate”**



“tidyverse” package

Here I introduce only some:

- “**ggplot2**”: *ggplot2* is a system for declaratively creating graphics, based on **The Grammar of Graphics**. You provide the data, tell *ggplot2* how to map variables to aesthetics, what graphical primitives to use, and it takes care of the details.
- “**dplyr**”: *dplyr* provides **a grammar of data manipulation**, providing a consistent set of verbs that solve the most common data manipulation challenges.
- “**readr**”: *readr* provides **a fast and friendly way to read rectangular data** (like csv, tsv, and fwf). It is designed to flexibly parse many types of data found in the wild, while still cleanly failing when data unexpectedly changes.

“tidyverse” package

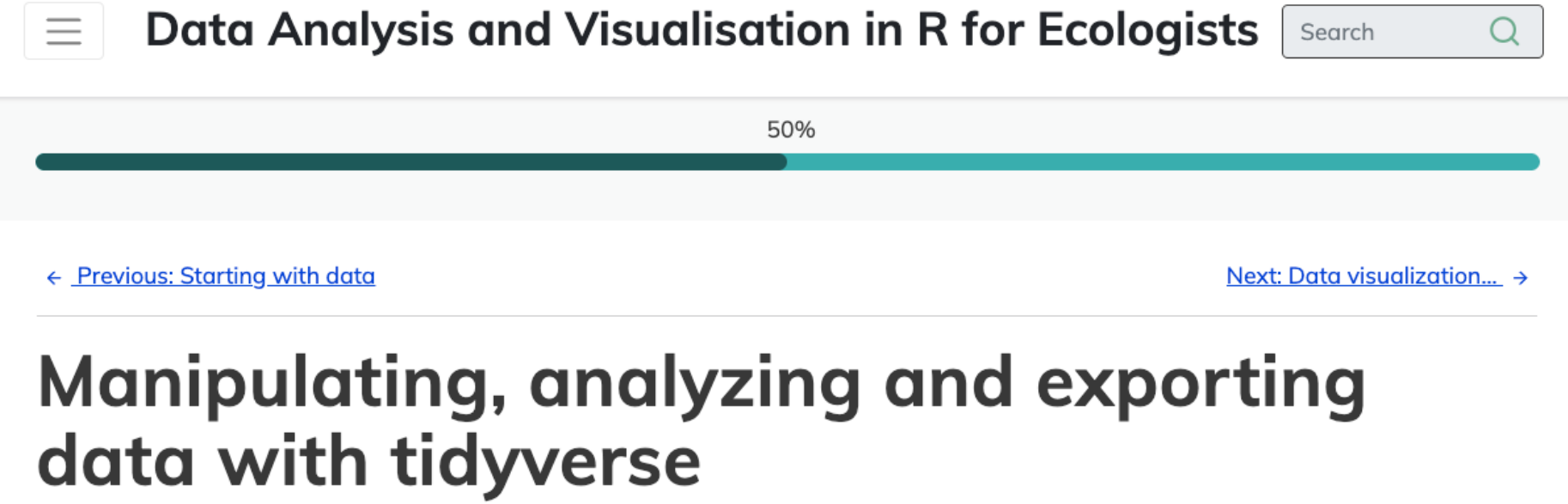
Here I introduce only some:

- “**lubridate**”: *Date-time data can be frustrating to work with in R.* Lubridate makes it easier to do the things R does with date-times and possible to do the things R does not.

Data manipulation with tidyverse

Data manipulation

- In the hands-on, we will use tidyverse functions to manipulate data.
- We will skip the data manipulation section in the Data Carpentry website due to time limitation (you can try if you have time. It is definitely useful!).
- If we come across code that I have not explained, I will explain it each time.



https://datacarpentry.org/R-ecology-lesson/03-dplyr.html

Data manipulation

Four points:

1. We will use **"tidyverse" functions** that were specifically designed to manipulate data.frame (or tibble).
2. However, you can do the same manipulation even if you use **"base" functions** only (You learned it in "*R Basics – Starting with data*"). The difference between these two is "**processing speed**" and "**readability**".
3. "**Tidy data**" is an important concept when you analyze and visualize data.
4. We will use "**pipes**" to process data.

Data manipulation

“pipe” (%>% or |>) :

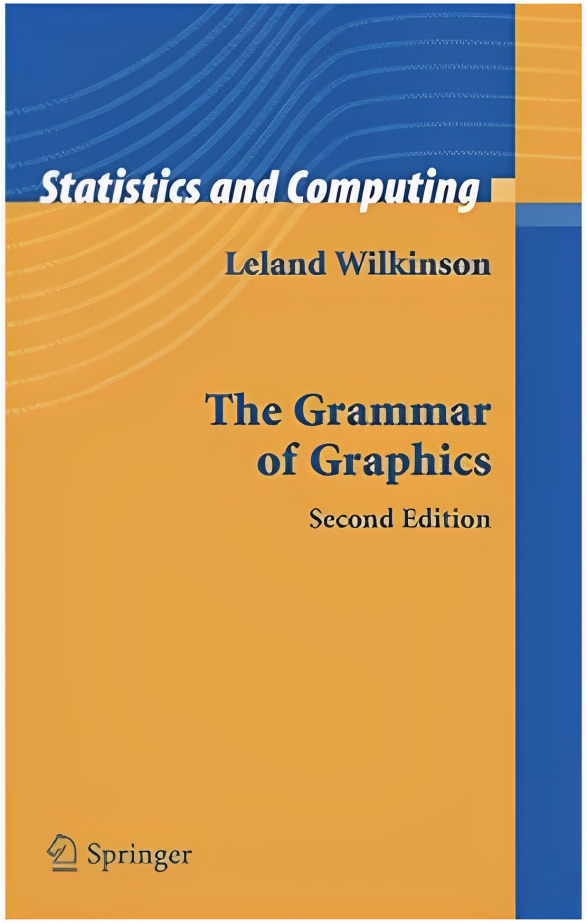
1. “pipes” are a recent addition to R.
2. Pipes let you take **the output of one function and send it directly to the next**, which is useful when you need to do many things to the same dataset.
3. Pipes in R look like **%>%** and are made available via the “**magrittr**” package, installed automatically with “dplyr”. If you use RStudio, you can type the pipe with Ctrl + Shift + M (windows), or Cmd + Shift + M (mac).
4. More recently, a new version of pipe, **|>**, was introduced to R4.1.0. However, we will **NOT** use this pipe. If you are interested, please read https://www.tidyverse.org/blog/2023/04/base-vs-magrittr-pipe/.

ggplot2 basics

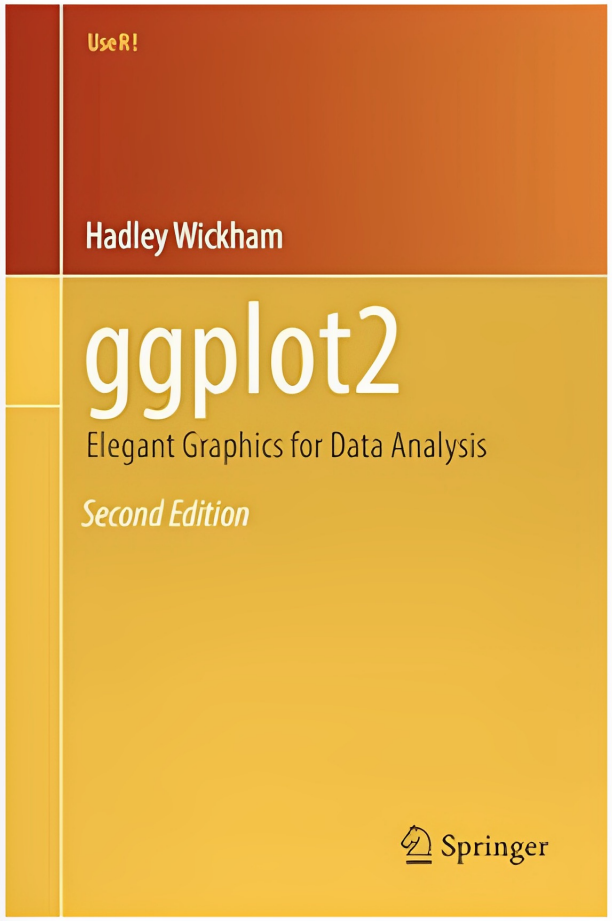
THE GRAMMAR OF GRAPHICS

“The Grammar of Graphics”

- “**The Grammer of Graphics**” presents a theoretical framework for graphics and serves as the basics of the ggplot2 package.
- According to this theory, a graph has a grammatical structure for visualizing data, and its elements (such as axes, labels, and legends) are considered as independent components.



Wilkinson (2005)
“The Grammar of Graphics”



Wickham (2016)
“ggplot2”

Idea behind ggplot2

- An object created by ggplot2 has a clear **structure**.

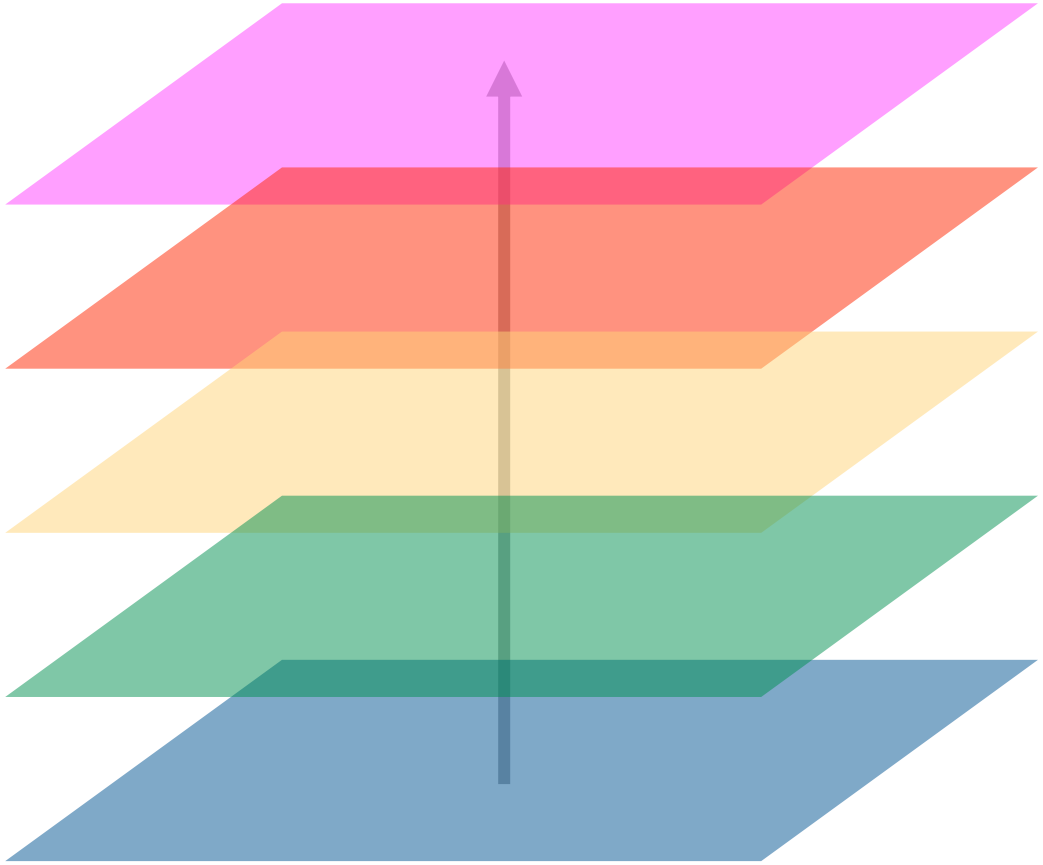
Statistics: Create multiple panels

Facets: Create multiple panels

Geometrics: Define plot type(s)

Aesthetics: Define “axes” and others

Data: Should be “tidy” data



Idea behind ggplot2

- An object created by ggplot2 has a clear **structure**.

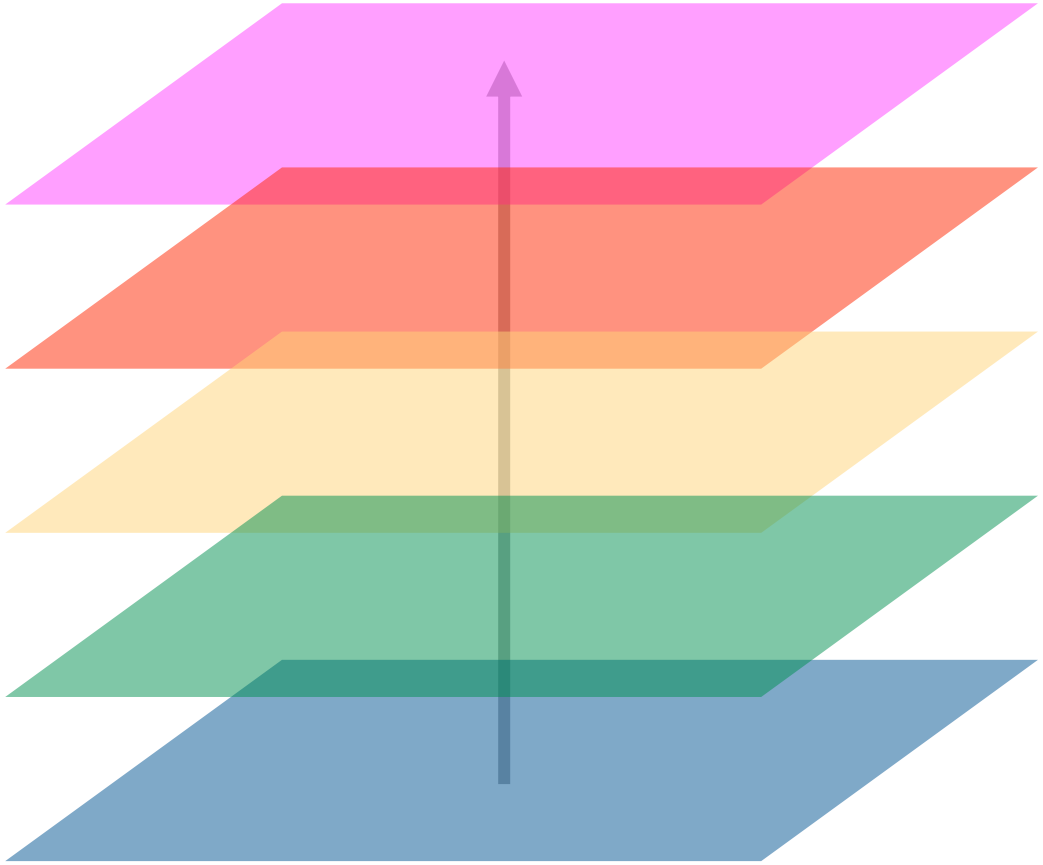
Statistics: `stat_xxxx()`

Facets: `facet_xxxx()`

Geometrics: `geom_xxxx()`

Aesthetics: `aes()`

Data: `ggplot(data= YOURDATA)`



Idea behind ggplot2

- An object created by ggplot2 has a clear **structure**.

Example 1:

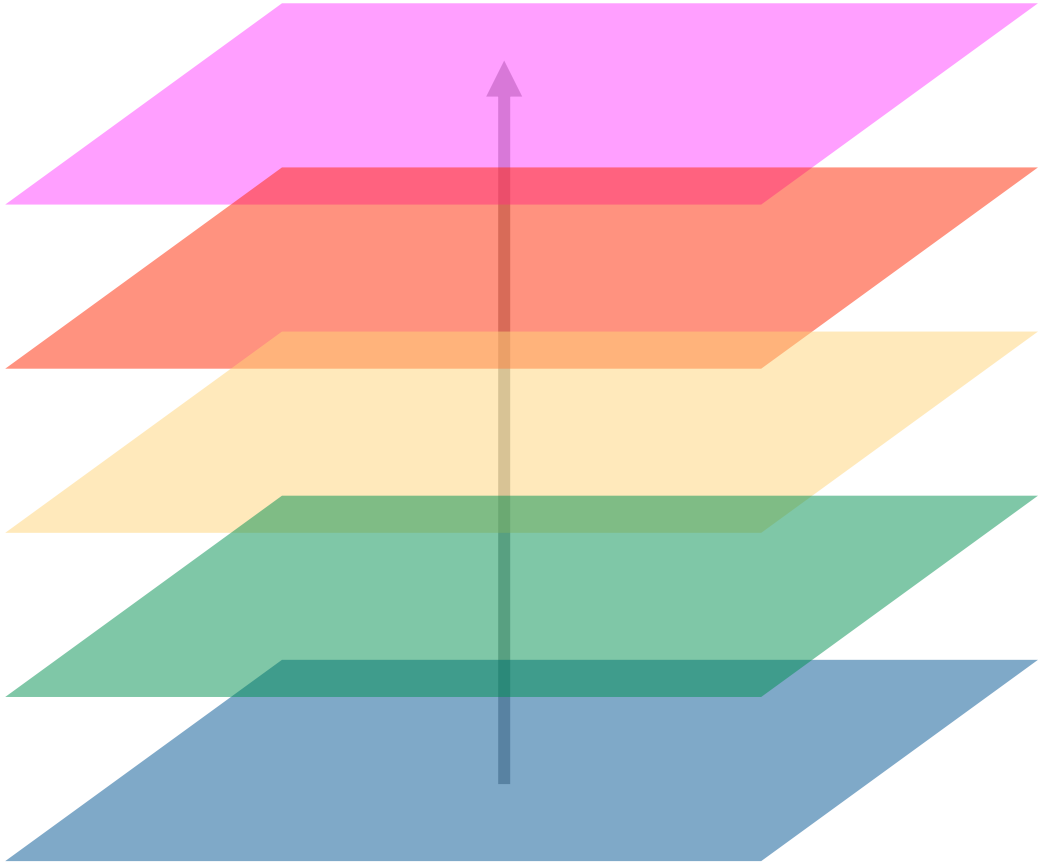
```
ggplot(data = d, aes(x = length, y = width)) +
     geom_point()
```

Example 2:

```
ggplot(data = d, aes(x = length, y = width)) +
     geom_point() +
     stat_smooth(method = “lm”)
```

Example 3:

```
ggplot(data = d) +
     geom_point(aes(x = length, y = width)) +
     facet_wrap(~ species) +
     stat_smooth(method = “lm”)
```



Short introduction to ggplot2 “Layers”

https://youtu.be/PiY9hwOkL8U?si=I2pBicMoTqdj_PIJ

ggplot2 & plotnine

ggplot2 is an R package (not Python) and is a system for declaratively creating graphics, based on *The Grammar of Graphics*.

plotnine is an implementation of the grammar of graphics in Python based on **ggplot2**.

ggplot2 3.4.4 Reference News Articles Extensions

ggplot2

plotnine 0.12.4 API Gallery Tutorials Site Page

A Grammar of Graphics for Python

https://ggplot2.tidyverse.org/ https://plotnine.readthedocs.io/en/v0.12.4/